Market Technicians Association, Inc.
Professionals Managing Market Risk • Incorporated in 1973
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Chartered Market Technician (CMT) Program – Level III

The CMT Level III exam will test the candidates' ability to **integrate** their understanding of concepts identified in Level I studies with the practical application learned in Level II studies. The Level III exam requires candidates to **implement** critical analysis to arrive at well-supported recommendations in an investing/trading context. A successful Level III candidate demonstrates they are ready to apply technical analysis in an institutional setting.

The CMT Level III exam is in short answer format and is computerized. Price information, data, and charts will be on screen and printed copies will also be provided. All candidates must be pass the Ethics portion of the exam to be successful.

Exam time length: 4 hours, 15 minutes
Exam format: Short Answer

The curriculum is organized into exam specific knowledge domains that provide a framework for recognizing and implementing investment/trading decisions. CMT Level III exam tests the candidate's knowledge in seven domains:

1. Risk Management
2. Asset Relationships
3. Portfolio Management
4. Classical Methods
5. Behavioral Finance
6. Volatility Analysis
7. Ethics

CMT®
CHARTERED MARKET TECHNICIAN

CMT Level III Exam Topics & Question Weightings

1. Risk Management	a.	risk management (e.g., basics of probability and statistics, basics of modeling risk factors, number of assets and impact on portfolio, managing risk through correlation, value-at-risk, performance and risk metrics, market volatility and fat-tailed distributions, correlation and diversification, managing individual trade risk, managing risk for an entire portfolio)	21%
	b.	position sizing	
	c.	quantitative and statistical analysis	
	d.	system development and testing	
2. Asset Relationships	a.	intermarket analysis	18%
	b.	relative strength	
	c.	sector rotation	
3. Portfolio Management	a.	portfolio management (performance measurement, portfolio allocation, asset correlation, asset allocation, alternative investments, risk management with alternatives)	18%
4. Classical Methods	a.	sentiment	21%
	b.	market breadth	
	c.	market forecasting	
	d.	price patterns	
	e.	volume study and analysis	
	f.	candlestick analysis	
	g.	oscillators or various technical studies	
5. Behavioral Finance	a.	behavioral finance	10%
6. Volatility Analysis	a.	volatility analysis	7%
7. Ethics	a.	ethics	5%

CMT Level III Exam - Learning Objectives

1. Risk Management	
Chapter References For This Domain: 1 – (Evaluation of) Triple Screen Trading System 2 – Spreads and Arbitrage (Risk Reduction in Spreads) 3 – Behavioral Techniques (Measuring the News, Event Trading) 5 – Day Trading (Intraday Breakouts, Price Shocks) 7 – Price Distribution Systems 9 – Advanced Techniques (Volatility, Liquidity, Trends and Price Noise, Trends and Interest Rate Carry) 10 – System Testing 11 – Practical Considerations (Use and Abuse of the Computer, Extreme Events, The Theory of Runs) 12 – Risk Control 22 – Portfolio Risk and Performance Attribution 23 – Hypothesis tests and confidence intervals 24 – Data Mining Bias: the fools gold of objective TA 58 – Statistics Summary for Chart Patterns 59 – Fact, Fiction, and Momentum investing	
basics of probability and statistics	Explain how to measure probability of price change and returns over a given time frame.
basics of measuring risk factors	Explain how to measure risk factors such as news, volatility, etc.
number of assets and impact on portfolio	Explain the impact of varying the number of assets and positions in a portfolio.
correlation and diversification	Analyze and explain the difference of risk between two different asset classes Critique diversification approaches based on correlations
value-at-risk	Interpret calculations of VaR Compare VaR calculation to confirm selection of stop placement
performance and risk metrics	Critique the use of performance and risk metrics based on a given objective
market volatility and fat-tailed distributions	Analyze fat-tailed distributions among returns data
managing individual trade risk	Calculate the amount of money at risk of being lost in a specified scenario.
managing risk for an entire portfolio	Calculate the amount of money at risk in a portfolio based on a specified scenario
position sizing	Explain how to change the risk in a scenario by adjusting the size of an investment position
quantitative and statistical analysis	Differentiate between random and nonrandom trends in data from system performance.
system development and testing	Interpret data from a system test to determine lack of randomness in the results

2. Asset Relationships	
Chapter References For This Domain: 3 – Behavioral Techniques (Commitment of Traders Report) 13 – Regression 14 – International Indices and Commodities 15 – The S&P 500 16 – European Indices 17 – Gold 18 – Intraday Correlations 19 – Intermarket Indicators 20 – Relative Strength (Relative Rotational Graphs)	
intermarket analysis	Analyze correlations between two or more asset classes Analyze and explain the difference of risk between two different asset classes.
sector rotation	Forecast possible progression of a business cycle model Explain the relationship between the business and financial cycles Identify leading, coincident and lagging indicators of economic activity
relative strength	Analyze and interpret relative strength of asset classes Analyze and interpret relative strength of Stock sectors Analyze and interpret relative strength of individual securities
3. Portfolio Management	
Chapter References For This Domain: 21 – Analyzing the Macro-Finance Environment (Leading, Coincident and Lagging Indicators) 22 – Portfolio Risk and Performance Attribution 32 – Hedging with VIX Derivatives 59 – Fact, Fiction, and Momentum investing	
performance measurement	Explain the differences of various performance metrics and why one is more suitable than another for a given objective. Interpret the Sharpe and Treynor ratios for individual stocks and portfolios.
tactical asset allocation	Evaluate performance data from returns generated by investment or trading.
asset correlation	Prepare a recommendation or other response based on asset correlation data
alternative investments	Explain the characteristics of different alternative investment types and why a portfolio manager might consider using them

4. Classical Methods	
Chapter References For This Domain: 3 – Behavioral Techniques (Commitment of Traders Report , Opinion and Contrary Opinion, Fibonacci and Human Behavior, Price Target Constructions Using the Fibonacci Ratio) 4 – Pattern Recognition 5 – Day Trading (Trading Using Price Patterns, Intraday Breakouts, Volume, Price Shocks) 6 – Adaptive Techniques 7 – Price Distribution Systems (Using Daily Distributions to Identify Support and Resistance) 8 – Multiple Time Frames 33-47 Candlestick Patterns 48-58 Price Patterns	
sentiment	Interpret investor sentiment from a COT data
market breadth	Identify and Interpret measures of market breadth
market forecasting	Forecast market trends and trend changes based on given charts displaying any combination of classical methods Forecast trends and potential trend changes in individual assets (stock prices, bond prices, commodity prices etc.)
price patterns	Evaluate and interpret price patterns Calculate potential price targets Evaluate price levels for potential support or resistance
volume study and analysis	Interpret volume signals Identify early warning signals from intraday volume
candlestick analysis	Identify and interpret candlestick patterns Validate a forecast with candlestick patterns
oscillators or various technical studies	Identify and Interpret signals from various oscillators and technical studies Identify and interpret divergence signals between two price series Identify and interpret divergence signals between a price series and an oscillator.
5. Behavioral Finance	
Chapter References For This Domain: 25 – Causality and Statistics 26 – Illusion 27 – The Story Is the Thing (The Allure of Growth) 28 – Are Two Heads Better than One? (Beating the Biases) 29 – The Anatomy of a Bubble 30 – De-bubbling: Alpha Generation	
	Discuss cognitive limitations investors often face Critique a given investment selection process for influence of behavioral biases Distinguish between cognitive errors and emotional biases Identify key points for exploiting a debubbling process

6. Volatility Analysis	
Chapter References For This Domain: 9 – Advanced Techniques (Volatility sections) 22 – Portfolio Risk and Performance Attribution 31 – The VIX as a Stock Market Indicator 32 – Hedging with VIX Derivatives	
	Contrast different measures of volatility Interpret changes in volatility as a signal useful for forecasting Explain how volatility can be an integral part of a market forecast Identify the subcomponents of portfolio volatility Explain how portfolio volatility may be affected by diversification
7. Ethics	Code of Ethics and Standards of Professional Conduct